

Ultrasonic Blood Flow Monitoring System with Threshold-Based Alerting

Details of Team Members

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ABSTRACT:

Ultrasonic blood flow measurement is a critical non-invasive technique widely used in biomedical diagnostics to assess cardiovascular health. This project presents the design and development of a smart ultrasonic blood flow monitoring system that not only measures blood velocity but also provides real-time threshold-based alerting for abnormal conditions. The system operates on the principle of the Doppler Effect, where high-frequency ultrasonic waves are transmitted into the blood vessel, and the frequency shift of the reflected signal from moving blood cells is analyzed to determine flow velocity. Using the received frequency from the artery, we implemented multiple buffers, along with a high pass filter to remove unstable DC offsets from biosignals, followed by a DC offset generator to produce a stable positive voltage reading. This sinusoidal signal was then converted to a square wave via an inverting hysteretic comparator which feeds the resulting square wave into an inverting amplifier to invert the negative voltage reading to a positive voltage reading. The processed data is fed into arduino, where digital signal processing algorithms compute the instantaneous blood flow velocity. The system continuously monitors this velocity and compares it against predefined physiological thresholds. If certain thresholds are met then alerts are given. Alerts are communicated through audible alarms. The benefit of this type of alarming is that the patients can get alerted even while sleeping. The proposed system offers advantages such as non-invasive operation, real-time monitoring, improved diagnostic capability, and automated anomaly detection. By combining ultrasonic sensing with intelligent instrumentation and alerting, this project demonstrates a practical and scalable solution for modern biomedical monitoring systems.

1)INTRODUCTION:

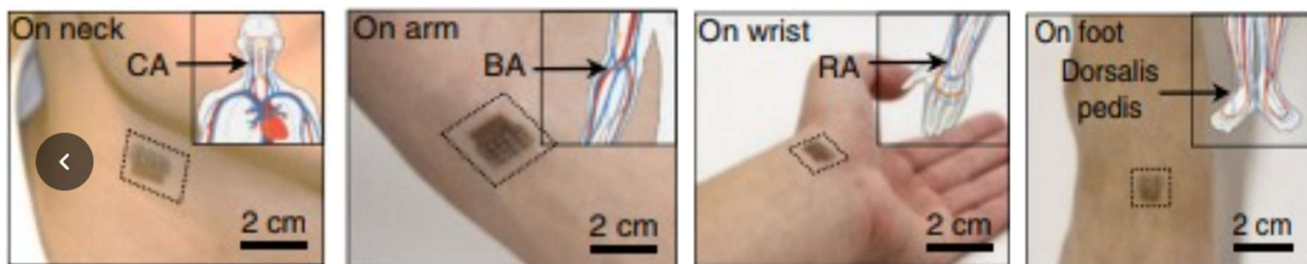
Measurement of blood flow velocity plays a vital role in the diagnosis and monitoring of cardiovascular conditions. Variations in blood flow can indicate serious health issues such as arterial blockages, hypertension, or poor circulation. Conventional methods of measuring blood flow are often invasive, expensive, and unsuitable for continuous monitoring, which limits their practicality in regular healthcare applications. To address these challenges, this project focuses on the development of a non-invasive ultrasonic blood flow monitoring system based on the principle of the Doppler Effect. In this technique, ultrasonic waves are transmitted into the blood vessel, and the frequency shift of the reflected waves caused by moving blood cells is used to determine the velocity of blood flow. This approach ensures safe, real-time, and accurate measurement without causing discomfort to the patient. The motivation behind this project arises from the increasing need for early detection and continuous monitoring of cardiovascular abnormalities. Many critical conditions develop gradually and remain undetected until they become severe. By continuously measuring blood flow and integrating an intelligent threshold-based alert system, the proposed device enhances its diagnostic capability.

For Example, the mechanism of alert system in brain and neck arteries is:

- 1) If blood flow velocity exceeds 40 cm/s, the system triggers a high-flow alert, indicating possible conditions such as hypertension or arterial narrowing.
- 2) If blood flow drops below 30 cm/s, a low-flow alert is generated, which may indicate poor circulation, blockage, or cardiac insufficiency.

Furthermore, this project highlights the application of modern measurement and instrumentation techniques in biomedical engineering. By combining ultrasonic sensing, signal processing, and embedded system design, it aims to develop a cost-effective, portable, and reliable solution that can be used not only in hospitals but also for remote or home-based health monitoring. In summary, the project is motivated by the need to transform a conventional measuring instrument into a smart healthcare system that not only measures physiological parameters but also assists in early diagnosis, continuous monitoring, and preventive healthcare.

Different positions at which we can use this flowmeter:



2)CURRENT STATUS OF THE SYSTEM:

Ultrasonic sensors blood flow meters are primarily used in medical diagnostics and surgical monitoring. They provide a non-invasive way to measure the velocity and volume of blood circulating through arteries and veins.

Some of the key medical applications are:

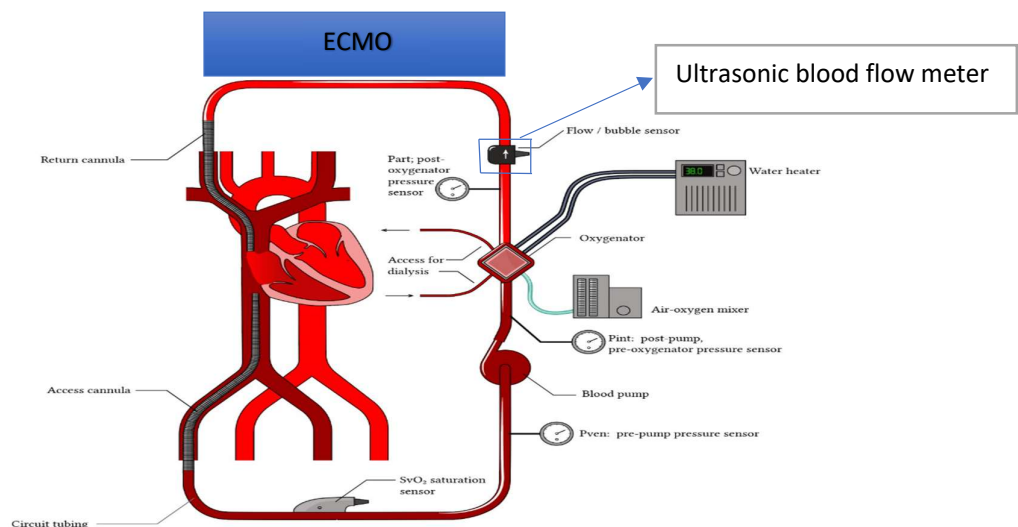
1)**Surgical Monitoring:** used in Intraoperative Monitoring, to know about Transplant success, Cerebrovascular Monitoring. Surgeons use them for real-time evaluation of blood vessels during procedures to ensure adequate blood supply to organs.

2)**Vascular Diagnostics:** Used in Echocardiography to access heart size, shape and pumping, to help diagnose **Deep Vein Thrombosis (DVT)**, **Peripheral Artery Disease (PAD)**, and carotid artery stenosis (narrowing) and used in aneurysm detection also.

3) **Patient Monitoring (ECMO):** In Extracorporeal Membrane Oxygenation (ECMO), ultrasonic flowmeters continuously measure the external flow of blood through the machine, ensuring the patient remains hemodynamically stable.

4) **Obstetrics and Foetal Health:** Ultrasound is the preferred method for monitoring prenatal development due to its safety for both mother and baby:

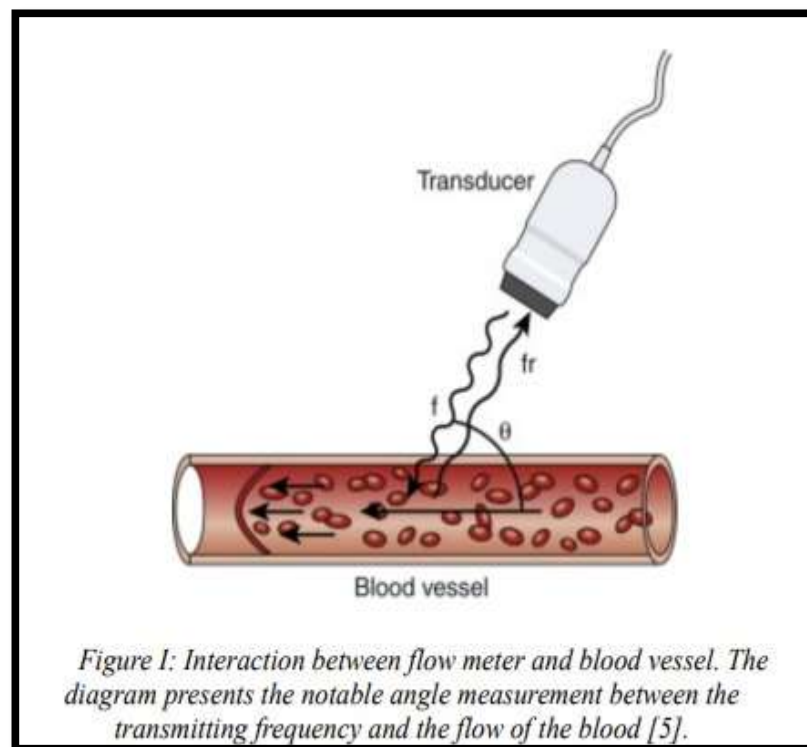
- **Foetal Well-being:** Clinicians evaluate blood flow in the umbilical cord, placenta, and fetal organs to ensure the baby receives adequate oxygen and nutrients.
- **Detecting Complications:** It helps identify Intrauterine Growth Restriction (IUGR), placental insufficiency, and foetal distress early, allowing for timely intervention.
- **Maternal Health:** Monitoring uterine artery flow can help predict and manage conditions like preeclampsia and gestational hypertension.



3)SYSTEM DESCRIPTION:

1)CONCEPTS:

A) Doppler effect: The Doppler effect describes the change in frequency of sound waves produced by a moving source with respect to an observer. Motion of the source relative to the observer changes the frequency of sound waves received by the observer. Ultrasonic flow meters obtain the Doppler frequency shift. These devices include a transmitter and receiver. Reflective material bounces the transmitted wave back into the receiver. The received wave has a frequency shift. The source frequency is lowered twice. The first shift occurs when the transmitted wave hits the object and the second shift occurs when the resulting wave comes back. This can be seen in Figure 1. The angle between the transmitted wave and flow factors into Doppler frequency shifts. In order for the flow meter to work, there must be moving solid particles or air bubbles present in the flowing liquid to reflect the ultrasonic beam. In our case, the transmitted high-frequency wave into the blood bounces off a red blood cell and back into the receiver. During this process, the source frequency is lowered twice: when the transmitted wave hits the object and when the reflected wave comes back to the receiver.



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Doppler frequency shift is linearly proportional to the velocity of the red blood cell as seen in the equation1:

$$\Delta f = f_r - f_t = \left(\frac{2v}{c} \right) * \cos(\theta) * f_t$$

Equation 1 [5]: Doppler frequency shift formula. θ is the angle of insonation (angle between the transmitting ultrasonic beam and the blood flow vector), v is the fluid flow velocity to be measured and displayed by the circuit, and c is the speed of sound in blood, ≈ 1578.2 m/s [13].

The flow meter should be oriented so that $\theta = 45^\circ$ to produce the most reliable results . This ensures the wave is neither too perpendicular (producing zero Doppler shift) nor too parallel (producing maximum Doppler shift) to the blood flow.

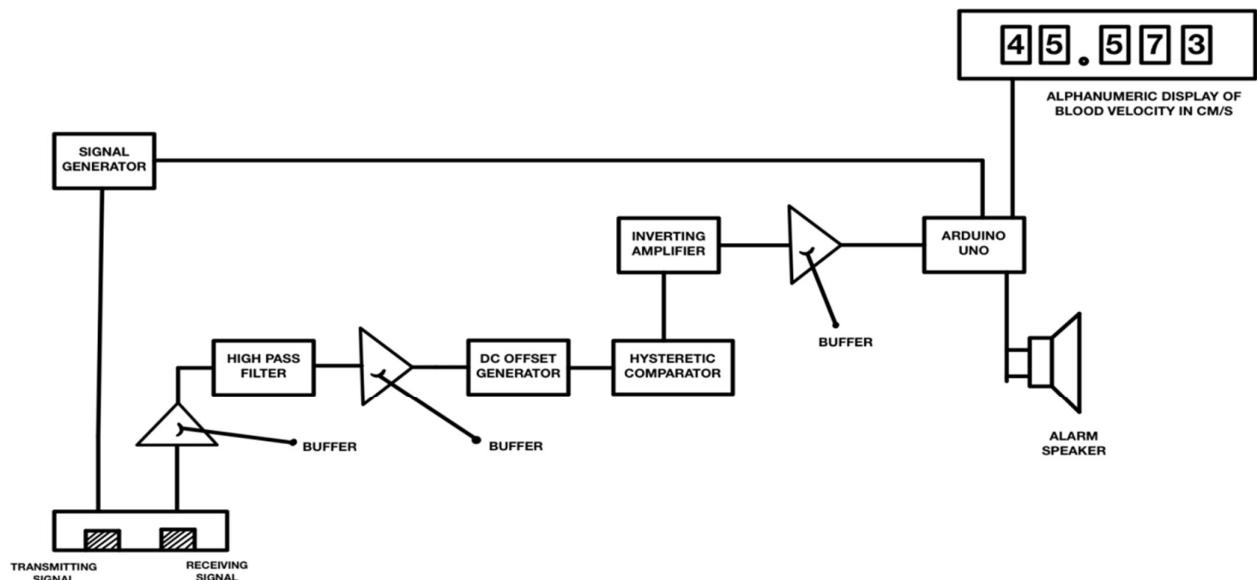
B) Fluid Dynamics: Our device relies upon detecting changes in blood velocity, which must increase at the site of the blockage due to Bernoulli's principle (Equation II), which states that fluid flow conserves energy.

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

Arterial blood flow, unlike vessel blood flow, is pulsatile rather than constant, so blood velocity follows a parabolic shape with each heartbeat. For the purposes of this study, we assume that arterial blood flow is laminar and follows a parabolic velocity profile, where maximum velocity is at the center and the mean velocity half of the maximum velocity. This allows us to assume that the blood flow vector is parallel to the arterial wall and that blood velocity is measured at the middle of the artery.

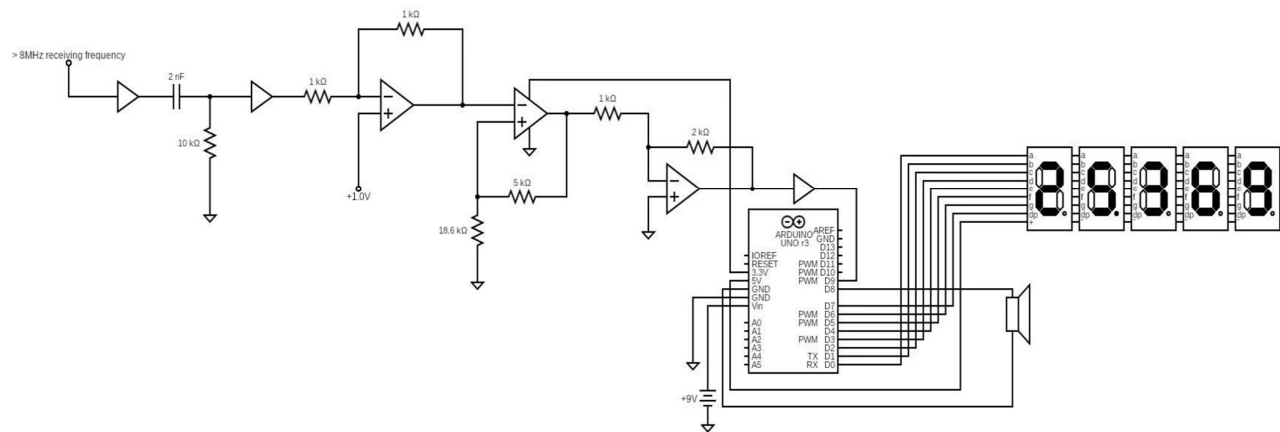
II)METHODS:

Simple Block Diagram of Ultrasonic Transducer Circuit:



In the above circuit, the signal is transmitted through transmitter. After reflecting from the blood cells, the reflected signal is received at receiver. The received signal will be passed through a series of 1) Buffers, 2) High Pass Filter, 3) DC Offset Generator, 4) Hysteretic Comparator, 5) Inverting Amplifier and fed into an 6) Arduino Uno for processing. The transmitted signal is also fed into the Arduino Uno for calculation of velocity.

Circuit Diagram of Ultrasonic Blood Flowmeter:



Explaining different blocks given in the block diagram:

- 1) **High Frequency Signal Generator:** The ultrasonic transducer consists of two piezoelectric crystals. One is for transmitting a stable and continuous 8 MHz frequency into the artery and the other is for receiving the reflected signal. Specifically, the high frequency signal is produced by the piezoelectric effect caused by the piezoelectric crystals within the transducer where the generated frequency depends on the thickness of piezoelectric crystal and the voltage that is applied to it.
- 2) **Buffer:** Three digital buffers were strategically placed throughout the circuit. The first buffer is placed at the beginning of the receiving circuit, the second buffer is placed at the end of the receiving circuit, before the Arduino Uno and the last buffer is placed after the high pass filter. The function of the first two digital buffers in our circuit is to isolate the ultrasonic transducer and the Arduino from the rest of the circuit. The third digital buffer that comes after the high pass filter was added to prevent the interference between the output signal of the high pass filter and the rest of the circuit. Our goal is to ensure that all signals can transition from one component or stage to another without interruption or interference due to the mismatched impedance of different components.
- 3) **High Pass Filter:** During the process of recording analog signals, bio signals are taken into consideration since it is not completely valid to assume that the only signal that is being picked up is the signal from the blood flow. With the presence of bio signals, it is expected that there will be a level of DC offset fluctuation. DC offset fluctuation can affect the circuit's analysing stage, which

ultimately alters the digital reading at the output. The low pass filter eliminates the DC offset fluctuation, as well as. enforces the signal to stabilize around the x-axis.

$$f_c = \frac{1}{2\pi RC}$$

The cut off frequency of the high pass filter was designed to be 8 kHz which can be achieved using a 2 nF capacitor in series with a 10 kΩ resistor.

- 4) **DC Offset Generator:** The purpose of incorporating a DC Offset Generator is to generate a stable DC offset which shifts the sinusoidal signal into the positive voltage region. It may seem counterintuitive that we eliminated DC offset with the high pass filter then added a DC Offset Generator. The reasoning behind this is the initial DC offset was unstable and was fluctuating between 0-5 V over time due to bio signals. With the DC Offset Generator, we are producing a stable and constant DC offset voltage which guarantees a consistent positive voltage reading, even over an extended period.

$$Gain = \frac{-R_f}{R_{in}}$$

The DC Offset voltage can be determined by the input voltage at the positive terminal of the Op-Amp. The DC Offset Generator also provides a signal gain but since the processing signal does not need to be amplified, Rf and Rin were chosen to be 1 kΩ each.

- 5) **Hysteretic Comparator:** A hysteretic comparator is a 1-bit analog-to-digital converter. With the input being a sinusoidal signal, the purpose of the comparator is to saturate the voltage as the input signal passes a certain threshold (Vref + and Vref -), forming a square wave as a result.

$$V_{ref}^+ = \frac{R1}{R1+R2} * V^+$$
$$V_{ref}^- = \frac{R1}{R1+R2} * V^-$$

We implemented a hysteretic comparator to convert the analog input into digital input which we can feed into the microcontroller as highs and lows. The value of R1 was chosen to be 18.6 kΩ and R2 to be 5 kΩ for the comparator to have threshold voltages of 2.4 V and 1.6 V.

- 6) **Inverting Amplifier:** Since the hysteretic comparator is an inverting comparator, the resulting square wave is generated in the negative voltage region. We introduced an inverting amplifier to reverse the negative voltage reading into positive voltage reading.

$$Gain = \frac{-R_f}{R_{in}}$$

The gain of the inverting amplifier is defined as the negative ratio between the input resistor (Rin) and the feedback resistor (Rf). By setting Rin = 1 kΩ and Rf = 2 kΩ, the signal was amplified by a factor of -2. This is a significant step in our circuit since we must consider that Arduino Uno only takes and reads positive voltage readings.

- 7) **Arduino:** An Arduino Uno is a type of microcontroller that we will be using to generate algorithms for performing arithmetic and logic operations. A recording for the alarm is stored within the Arduino which will be used as an audio output for the alarm speaker. An alphanumeric display is connected to the digital output ports (D0 – D8), on the Arduino Uno, for displaying a live reading for the blood flow velocity. The Arduino will also be powering the entire circuit via an external 9 V battery connected to the Vin port. The 5 V and 3.3 V ports are used to power the op amp of the comparator, the alphanumeric display, and the speaker.

III)SOURCES OF ERROR:

One potential source of error arises in the physical properties of the blood, including the concentration and size of blood cells. If the concentration of blood cells in the artery is too low, the accuracy of the device will go down since it depends on the presence of blood cells . Hence, it is not advisable to use this device on patients with severe anaemia. Moreover, the sonic conductivity of the blood, affected by blood cell physiological properties, is important. If the blood cells have low sonic conductivity, there will be inaccurate measurements since the ultrasonic signal is not able to properly bounce off the cells back into the transducer . This can create cascading effects in the blood velocity calculations. Flow meters are sensitive to disturbances in flow - the propagation of a sound wave may be impacted by turbulent flow, resulting in an incorrect measurement of velocity due to the presence of irregular streamlines with varying velocities. Furthermore, there are several sources of noise in the system above. Noise may arise from movements in the walls of the artery itself. However, the use of a high pass filter eliminates the likelihood of this impeding accuracy.

IV)ASSUMPTIONS:

One assumption made is that sound waves travel in a straight path from the transmitter to the artery and back to the transmitter, and that there is no interference between these two signals. Furthermore, there is an unstable DC offset due to bio signal fluctuation that occurs in the body which is in the range of 0-5 V which is addressed within our circuit design using filters and offset generators. We also assume that the blood flow direction is in the same direction as the artery, but this is not always true - in eccentric stenosis, flow direction and vessel direction may be different. This would make the assumed angle θ to be invalid. In the circuit shown above, all components are assumed to perform ideally.

4)OUR IDEA/VIEWS FROM DIFFERENT TEAM MEMBERS:

a) Transducer Engineer:

- I have researched and found that exactly a 45° angle between transmitted wave and blood flow to produce the most reliable Doppler shift results.
- I have also found that the required frequency of transmitted wave is 8MHz and how to generate it using piezoelectric devices.

b) Signal Conditioning Engineer:

- I have placed digital buffers strategically throughout the circuit to isolate components and prevent interference from mismatched impedance.
- I have also added High Pass Filters for noise filtering to remove unstable DC offsets and noise caused by natural bio signal fluctuations.
- I also added a DC offset generator in the circuit to produce a stable positive voltage reading. Following this, used an inverting hysteretic comparator to saturate the voltage and formed a square wave, and an inverting amplifier to reverse the resulting negative voltage into a positive voltage reading that the microcontroller can safely read.

c) Calibration & Standards Engineer:

- I determined the precise resistor values required for the circuit, such as setting the hysteretic comparator threshold voltages to 2.4 V and 1.6 V by selecting 18.6 k Ω and 5 k Ω resistors.
- I defined the standard operating parameters for the device, specifically setting the safety boundaries. For example, to alert users if blood velocity drops below 30 cm/s or rises beyond 40 cm/s in brain and neck arteries.

d) Error & Validation Engineer:

- I used software tools, such as the Falstad Java applet, to simulate the electronic circuit and verify whether the circuit is properly working or not.
- I analysed physical limitations that could cause errors, such as low blood cell concentration (e.g., severe anaemia) or low sonic conductivity, which would prevent the ultrasonic signal from bouncing back properly.

e) Data Acquisition Engineer:

- I have utilized an Arduino Uno to process the received frequency signals as digital highs and lows. I have programmed the arithmetic algorithms that calculate the actual fluid flow velocity using the Doppler frequency shift formula.
- I have connected an alarm to the Arduino and ensured that the audio alarm only triggers if the dangerous flow rate is sustained for more than 30 seconds.

5) CONCLUSION AND FUTURE DIRECTION:

Summary of the Proposed Device The proposed non-invasive ultrasonic blood flowmeter presents a highly viable, low-cost solution for the continuous monitoring and early detection of cerebrovascular insufficiency. By leveraging the Doppler effect and principles of fluid dynamics, the device successfully correlates frequency shifts in reflected ultrasonic waves to the continuous velocity of red blood cells within the carotid artery.

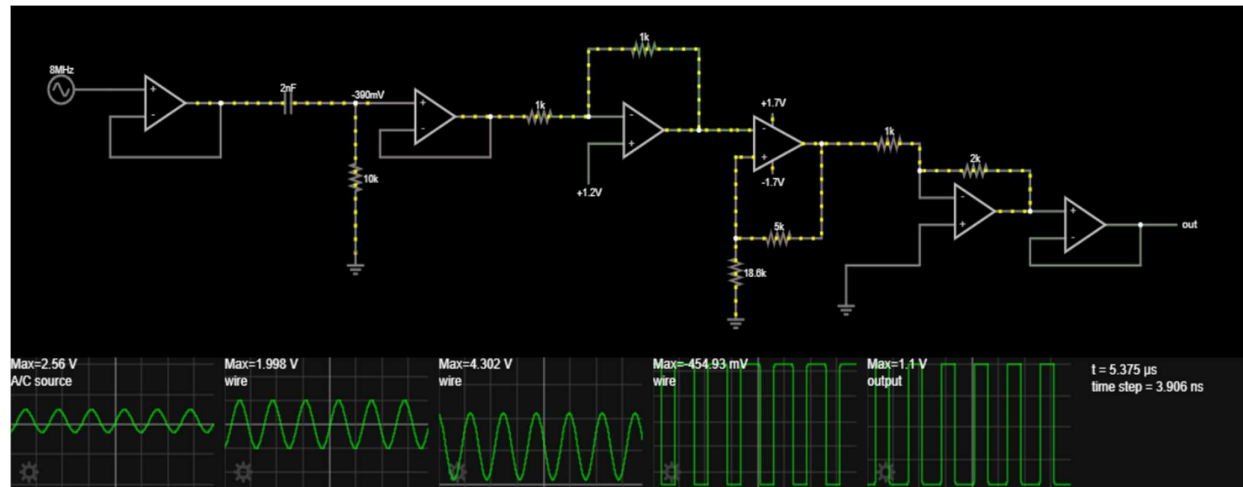
Technical Efficacy and Signal Processing The electrical design demonstrates a robust approach to mitigating common physiological and mechanical interference. The implementation of an 8 kHz high-pass filter effectively eliminates the unstable, low-frequency DC offsets inherent to natural bio signals and arterial wall movements. Furthermore, the subsequent signal processing chain—utilizing a DC offset generator, an inverting hysteretic comparator, and an inverting amplifier—ensures that the analog bio signal is accurately saturated into a stable, positive digital square wave. This critical conditioning allows the Arduino microcontroller to reliably process the data and trigger a life-saving auditory alarm if blood velocities fall below 30 cm/s or exceed 40 cm/s, warning of impending ischemic events before irreversible brain tissue necrosis can occur.

Clinical Limitations While the current model establishes a strong foundational framework, its accuracy relies on several ideal assumptions. Primarily, the system assumes purely laminar blood flow and a constant angle of insonation parallel to the arterial wall. In reality, particularly in cases of arteries exhibit irregular geometries and turbulent flow that can scatter ultrasonic propagation and yield inaccurate velocity calculations. Additionally, the device's reliance on ultrasonic reflection means its efficacy is highly dependent on the patient's blood cell concentration and sonic conductivity, rendering it less effective for patients with severe anaemia.

Future Outlook and Improvements Future iterations of this design have significant potential for clinical expansion. The implementation of variable frequency generators could allow the device to scan at different penetration depths, enabling the detection of turbulence and facilitating the surveillance of other critical vessels. To correct for insonation angle errors, a dual-probe system could be integrated to send simultaneous signals at opposing angles. Furthermore, transitioning the primary frequency generation from standard piezoelectric crystals to crystal oscillators offers an avenue to further reduce manufacturing costs without sacrificing signal stability. Ultimately, integrated into a wearable cervical foam collar, this device provides a crucial layer of passive, continuous surveillance, granting vulnerable patients the ability to sleep safely while monitoring for critical hemodynamic shifts.

6)GRAPHS AND CODES:

The fig below is Falstad Circuit Diagram of Ultrasonic Blood Flowmeter with a mimicked signal generator:



Arduino Pseudocode:

1. Declare variables a, ft, c, fr, θ , v
2. Transmitting frequency signal is recorded from the signal generator and initialized
3. The receiving frequency signal is processed, recorded and stored by the Arduino.
4. Calculate velocity using Doppler Effect a. $v = \Delta f * c / (2 * \cos(\theta) * ft)$
5. Store alarm audio file into the Arduino a. Recording is stored for alarm speaker
6. Send velocity values to the alphanumeric display
7. If $30 < \text{velocity} < 40$, the timer doesn't start → no alarm
8. If $\text{velocity} > 40$ or $\text{velocity} < 30$, the timer starts (0 to 30 seconds). If time > 30 seconds, trigger the alarm.
9. If statement for when button is pushed, Arduino turns off alarm audio recording

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- 3) Ultrasonic Doppler and Time of Flight Velocity and Volume Flow Meters." Engineering Toolbox, 2003,
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